

Slide 1: Query Execution Steps

23IT201 • Database Management Systems • Module 10: Query Processing & Optimization

High-level SQL query -> Parsing & Translation -> Relational Algebra Expression -> Query Optimizer -> Execution Plan -> Database Engine.

Slide 2: Cost-Based Query Optimizer

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Estimates CPU and disk I/O cost using database statistics (catalog histograms, tuple counts, block counts).

Slide 3: Join Execution Algorithms

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Nested Loop Join, Block Nested Loop Join, Hash Join, and Sort-Merge Join.